# Held-out prediction error and calibration by marker

Per-slide P99 normalization; stride-32 training; expression-weighted MLP

Normalized MAE (lower is better)

Bias: predicted − measured (ideal = 0)

PODXL, ZO1, CD10, VIM, PDPN, CAV, PCNA, NES, PanCK, CDH1, CD34, UMOD, CD31, CD138, CTNNB1, BCL2, IFNA6, CD4, aSMA, DAPI, CDK1, PTEN, STAT1, AQP2, CD93, HIF1a, GATA3, ATM, PARP1, Cd11c, CD8, COLIV, CD45, Ki67, NF2, HIF2a, CAIX, POSTN, CD14, PAX2, CD1a, IL6

0.00 0.05 0.10 0.15 0.20

Per-slide P99 normalized intensity units

0.00 0.02 0.04 0.06 0.08 0.10 0.12 0.14

Per-slide P99 normalized intensity units

42 markers; 39 patients. Equal-patient means of within-patient OOF metrics.
Bars: reported 95% patient-bootstrap CIs, conditional on trained predictions; shared-fold training uncertainty is not fully captured.
Marker order matches the correlation figure. MAE units differ from historical raw-intensity experiments.